

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change, Update]	178068	49.232079
1	[Git, Change, Create]	52544	14.527318
2	[Git, Change, Delete]	27895	7.712384
3	[Java, ByteCode, Member, Method]	13490	3.729703
4	[Java, ByteCode, Parameter]	13293	3.675237
5	[Git, Commit]	13289	3.674131
6	[File, Git]	11096	3.067812
7	[Git, Change, Rename]	10752	2.972703
8	[Java, ByteCode, Bound, ParameterizedType]	7279	2.012491
9	[Java, ByteCode, Bound]	7240	2.001709
10	[Java, ByteCode, Member, Field]	3609	0.997813
11	[Java, ByteCode, Bound, WildcardType]	2947	0.814784
12	[Java, Value, ByteCode, Annotation]	2931	0.810360
13	[Xml, Element]	2162	0.597748
14	[Java, ByteCode, Member, Method, Constructor]	2117	0.585306
15	[Xml, Text]	1450	0.400895
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.307168
17	[Java, ByteCode, Member, Method, Lambda]	968	0.267632
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.245513
19	[Type, File, Java, ByteCode, Class]	846	0.233901
20	[Json, Key]	702	0.194088
21	[Value, Json, Scalar]	685	0.189388
22	[Java, Value, ByteCode, Primitive]	672	0.185794
23	[Type, File, Java, ByteCode, JavaType]	649	0.179435
24	[Java, ByteCode, Member, Method, GenericDeclar...	578	0.159805
25	[Type, File, Java, ByteCode, ExternalType]	411	0.113633
26	[Git, Change, Copy]	346	0.095662
27	[Author, Git, Person]	299	0.082667
28	[Value, Array]	287	0.079349
29	[Committer, Git, Person]	252	0.069673

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

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Nodes per label combination (more than 0.5% overall)

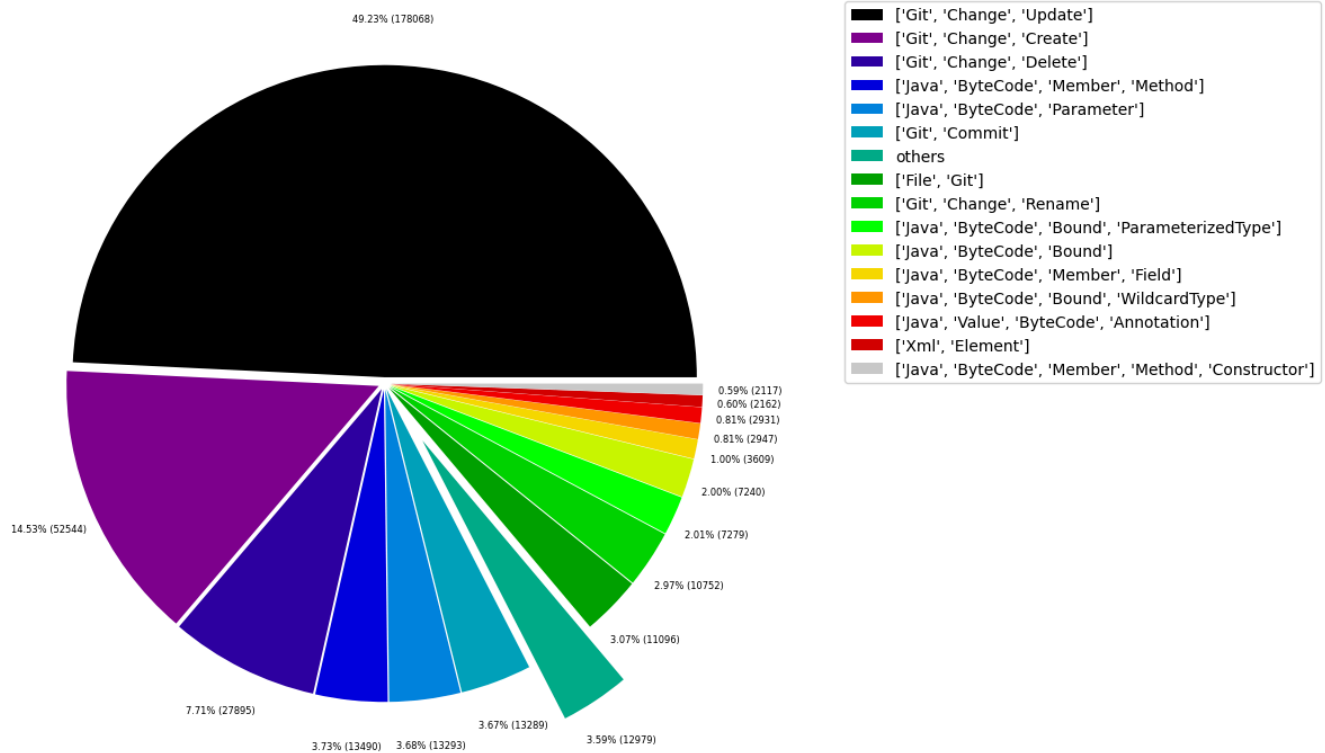


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000276
1	[Package, File, Json, NPM]	1	0.000276
2	[Repository, File, Git]	1	0.000276
3	[File, TS, Scan]	1	0.000276
4	[File, Json]	2	0.000553
5	[File]	3	0.000829
6	[Java, ByteCode, Member, Method, GenericDeclar...	4	0.001106
7	[Maven, Exclusion]	5	0.001382
8	[Value, Array, Json]	6	0.001659
9	[Dependency, NPM]	7	0.001935
10	[Type, File, Java, ByteCode, Void]	9	0.002488
11	[File, Maven, Xml, Pom, Document]	9	0.002488
12	[Java, ManifestSection]	9	0.002488
13	[File, Java, Manifest]	9	0.002488
14	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002488
15	[File, Java, ServiceLoader]	10	0.002765
16	[File, Java, Properties]	12	0.003318
17	[Maven, ExecutionGoal]	16	0.004424
18	[Maven, PluginExecution]	16	0.004424
19	[Xml, Attribute]	18	0.004977
20	[jQAssistant, Rule, Concept]	19	0.005253
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005806
22	[Maven, Configuration]	21	0.005806
23	[Maven, Plugin]	21	0.005806
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006083
25	[Type, File, Java, ByteCode, Enum]	28	0.007741
26	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008294
27	[Xml, Namespace]	36	0.009953
28	[Type, File, Java, ByteCode, Annotation]	44	0.012165
29	[Git, Branch]	47	0.012995

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

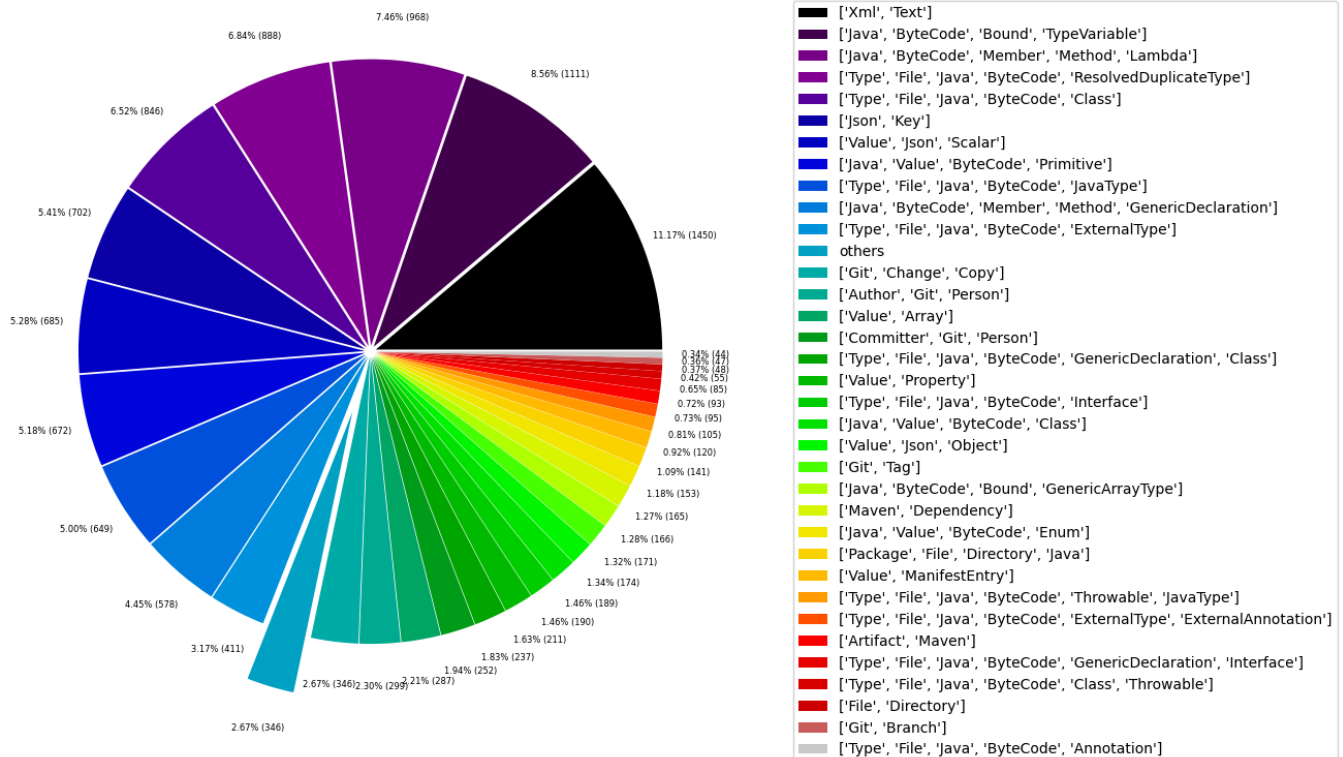


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

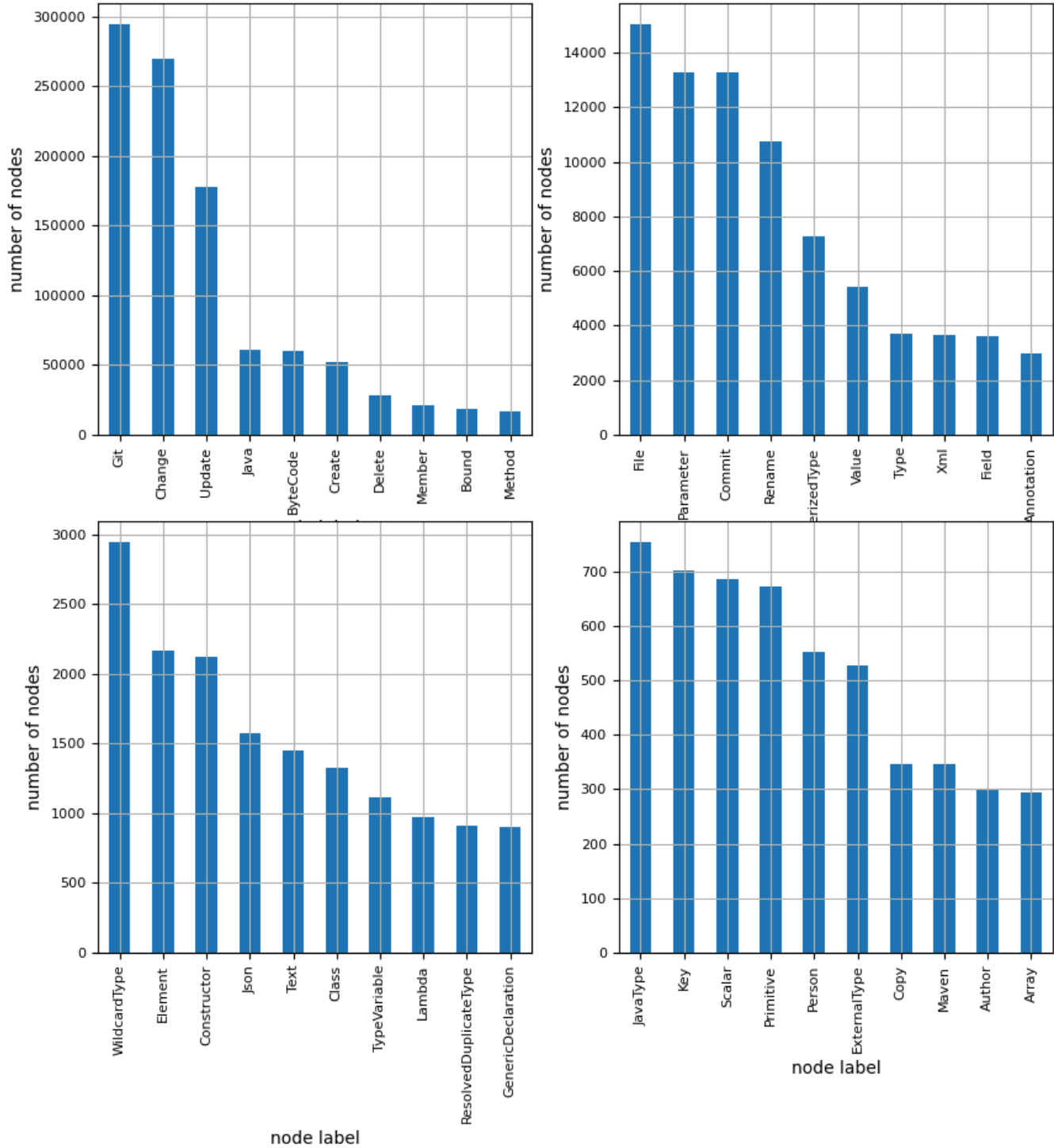
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	294760	81.494978
1	Change	269605	74.540146
2	Update	178068	49.232079
3	Java	60652	16.769010
4	ByteCode	60462	16.716479
5	Create	52544	14.527318
6	Delete	27895	7.712384
7	Member	20766	5.741365
8	Bound	18743	5.182048
9	Method	17157	4.743552
10	File	15057	4.162946
11	Parameter	13293	3.675237
12	Commit	13289	3.674131
13	Rename	10752	2.972703
14	ParameterizedType	7279	2.012491
15	Value	5428	1.500729
16	Type	3715	1.027120
17	Xml	3675	1.016061
18	Field	3609	0.997813
19	Annotation	2975	0.822525
20	WildcardType	2947	0.814784
21	Element	2162	0.597748
22	Constructor	2121	0.586412
23	Json	1570	0.434072
24	Text	1450	0.400895
25	Class	1327	0.366888
26	TypeVariable	1111	0.307168
27	Lambda	968	0.267632
28	ResolvedDuplicateType	910	0.251596
29	GenericDeclaration	904	0.249937
30	JavaType	754	0.208465
31	Key	702	0.194088
32	Scalar	685	0.189388
33	Primitive	672	0.185794
34	Person	551	0.152340
35	ExternalType	527	0.145704
36	Copy	346	0.095662
37	Maven	346	0.095662
38	Author	299	0.082667
39	Array	293	0.081008

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1122093

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	269605	24.026975
1	MODIFIES	269605	24.026975
2	UPDATES	178068	15.869273
3	CREATES	63642	5.671722
4	DELETES	38647	3.444189
5	INVOKES	36555	3.257751
6	COMMITTED	26578	2.368609
7	DEPENDS_ON	22492	2.004468
8	OF_TYPE	21920	1.953492
9	DECLARES	21227	1.891733
10	OF_RAW_TYPE	17353	1.546485
11	HAS_PARENT	16074	1.432502
12	HAS	14429	1.285901
13	HAS_COMMIT	13289	1.184305
14	RETURNS	12844	1.144647
15	HAS_FILE	11096	0.988866
16	RENAMES	10752	0.958209
17	READS	9378	0.835760
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.749225
19	HAS_NEW_NAME	6310	0.562342
20	OF_GENERIC_TYPE	5996	0.534359
21	RESOLVES_TO	5258	0.468589
22	SIMILAR	4078	0.363428
23	WRITES	3944	0.351486
24	CONTAINS	3936	0.350773
25	RETURNS_GENERIC	3593	0.320205
26	ANNOTATED_BY	2919	0.260139
27	REQUIRES	2230	0.198736
28	HAS_FIRST_CHILD	2162	0.192676
29	HAS_LAST_CHILD	2162	0.192676

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

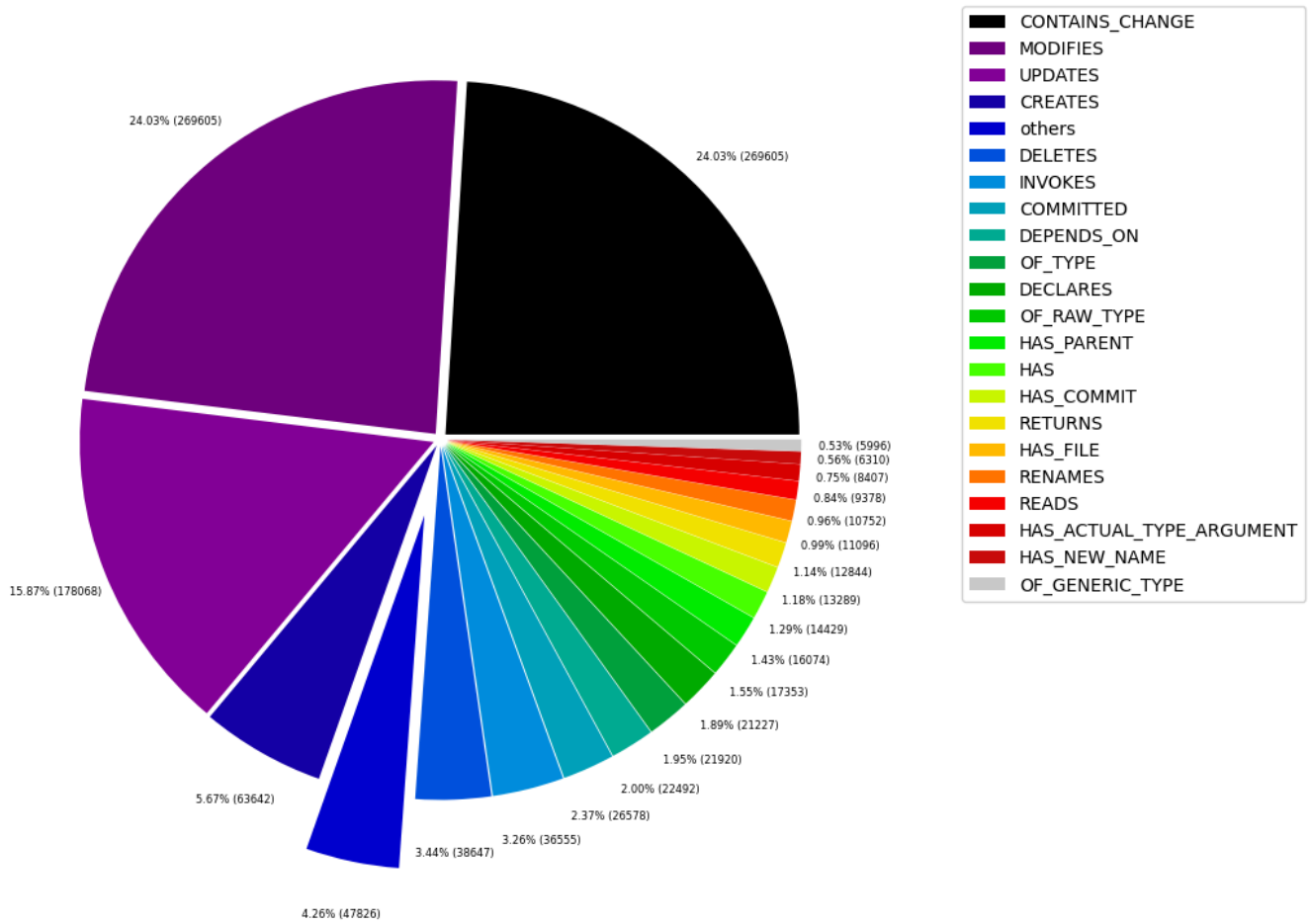


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

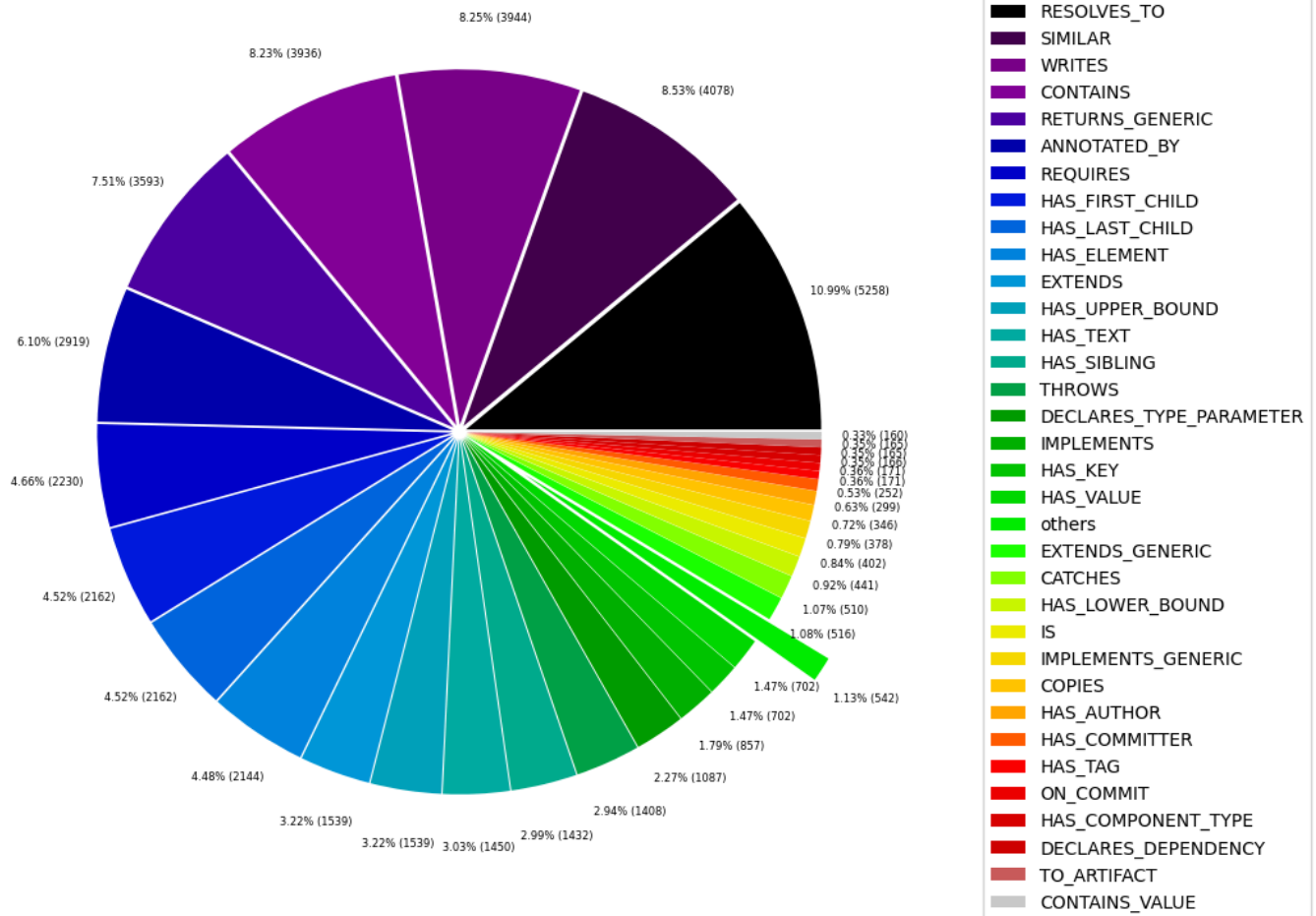
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000089
1	THROWS_GENERIC	5	0.000446
2	EXCLUDES	5	0.000446
3	DECLARES_DEV_DEPENDENCY	7	0.000624
4	DESCRIBES	9	0.000802
5	HAS_ROOT_ELEMENT	11	0.000980
6	HAS_GOAL	16	0.001426
7	HAS_EXECUTION	16	0.001426
8	OF_NAMESPACE	18	0.001604
9	HAS_ATTRIBUTE	18	0.001604
10	INCLUDES_CONCEPT	19	0.001693
11	IS_ARTIFACT	21	0.001872
12	HAS_CONFIGURATION	21	0.001872
13	USES_PLUGIN	21	0.001872
14	REQUIRES_TYPE_PARAMETER	24	0.002139
15	REQUIRES_CONCEPT	28	0.002495
16	DECLARES_NAMESPACE	36	0.003208
17	HAS_DEFAULT	39	0.003476
18	HAS_BRANCH	47	0.004189
19	HAS_HEAD	48	0.004278
20	COPY_OF	132	0.011764
21	CONTAINS_VALUE	160	0.014259
22	DECLARES_DEPENDENCY	165	0.014705
23	TO_ARTIFACT	165	0.014705
24	HAS_COMPONENT_TYPE	166	0.014794
25	ON_COMMIT	171	0.015239
26	HAS_TAG	171	0.015239
27	HAS_COMMITTER	252	0.022458
28	HAS_AUTHOR	299	0.026647
29	COPIES	346	0.030835

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Change, Update]	UPDATES	[File, Git]	178068		178068
1	[Git, Change, Update]	MODIFIES	[File, Git]	178068		178068
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Update]	178068		13289
3	[Git, Change, Create]	CREATES	[File, Git]	52544		52544
4	[Git, Change, Create]	MODIFIES	[File, Git]	52544		52544
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	52544		13289
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	27895		13289
7	[Git, Change, Delete]	DELETES	[File, Git]	27895		27895
8	[Git, Change, Delete]	MODIFIES	[File, Git]	27895		27895
9	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22326		13397
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	16065		13289
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	13289		1
12	[Committer, Git, Person]	COMMITTED	[Git, Commit]	13289		252
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	13289		299
14	[Repository, File, Git]	HAS_FILE	[File, Git]	11096		1
15	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	10752		13289
16	[Git, Change, Rename]	RENAMES	[File, Git]	10752		10752
17	[Git, Change, Rename]	DELETES	[File, Git]	10752		10752
18	[Git, Change, Rename]	CREATES	[File, Git]	10752		10752
19	[Git, Change, Rename]	MODIFIES	[File, Git]	10752		10752
20	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8515		13397
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8441		13397
22	[File, Git]	HAS_NEW_NAME	[File, Git]	6310		11096
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156		13293
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465		7240
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110		7279
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947		7279
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695		13293
28	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2519		2117
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2474		7279

Graph Density

```
total_number_of_nodes (vertices): 361691
total_number_of_relationships (edges): 1122093
-> total directed graph density: 8.577379967959107e-06
-> total directed graph density in percent: 0.0008577379967959107
```